Six Matches and a Wider Wedge

A sixth-order comparison for the Jensen polynomials of Riemann's xi-function

John Savva — Independent researcher

Version 1.1 — 3 September 2026

AI-assistance disclosure. These papers and the accompanying Lean 4 formalization were developed with substantial AI assistance, primarily from **OpenAI GPT-5.6 Sol Pro**. Additional models assisted with adversarial testing and AI-only technical review. **Wolfram Mathematica 15.0.1**, executed by the author, supplied clean-room computational corroboration of selected symbolic identities and exact algebraic calculations. AI systems are not authors, and no human mathematical review or peer review is claimed.

Version 1.1 is an attribution-only revision. The article body and mathematical claims are unchanged from Version 1.0.

MATHEMATICS

Six Matches and a Wider Wedge

A sixth-order comparison widens a known real-rootedness region for the Jensen polynomials of Riemann's xi-function. The argument also illustrates both the reach and the limits of formal proof in contemporary mathematics.

BY JOHN SAVVA · INDEPENDENT RESEARCHER · 21 AUGUST 2026

In 1914 the Danish mathematician Johan Jensen described a way to study an infinite analytic object through a sequence of finite polynomials. More than a century later, that idea has become one of the useful approaches to Riemann's xi-function, the symmetric form of the zeta function at the center of the Riemann hypothesis.

The finite objects are called Jensen polynomials. Each one is assembled from a short run of Taylor coefficients of xi. Their roots can be calculated, plotted and compared with the roots of classical polynomials. More importantly, their root geometry can sometimes be proved uniformly: not just for one degree or one starting coefficient, but throughout an expanding region of the two parameters.

The result described here is a proof candidate for a wider such region. In simplified asymptotic terms, an earlier growing-degree method allowed the degree to increase approximately as the three-fifths power of the coefficient index. The new argument permits growth at the two-thirds power, up to a logarithmic factor. The change comes from forcing a tractable comparison polynomial to agree with the xi polynomial through six consecutive coefficient conditions rather than five.

This is not a proof of the Riemann hypothesis. It is a theorem about a particular family of finite polynomials associated with xi. Its interest is threefold: it sharpens a growing-degree real-rootedness result, it exposes a specific sixth-order mechanism behind the new exponent, and it has been carried unusually far into machine-checkable form.

The last point requires care. A substantial new chain is checked by Lean, a formal proof assistant, and decisive algebra has been reconstructed in two independent computer-algebra systems. But the work has not received human mathematical review or peer review. It should therefore be read as an extensively checked, substantially formalized proof candidate whose detailed argument and verification record are open to inspection.

From zeta to an even entire function

For real numbers greater than one, the zeta function begins with the familiar series

$$\zeta(s) = 1 + 1/2^s + 1/3^s + 1/4^s + \cdots .$$

Analytic continuation extends zeta to almost the entire complex plane. Its nontrivial zeros carry information about the fine distribution of prime numbers, and the Riemann hypothesis asserts that every one of those zeros has real part one-half.

The raw zeta function is not the most convenient form for the present problem. After adjoining gamma and elementary factors, one obtains Riemann's completed xi-function. It is entire-there are no poles-and satisfies the functional equation $\xi(s) = \xi(1-s)$. Centering at one-half therefore produces an even function. With the normalization used in the paper, it has an expansion of the form

$$\xi(1/2 + z) = \Sigma\ \gamma(n)\ z^{2n}/n!.$$

The positive numbers $\gamma(n)$ are moments of an explicit, rapidly decaying theta kernel. They are the data from which the Jensen polynomials are built. For a starting index n and degree d, define

$$J^{d,n}(X) = \Sigma_{k=0}^{d}\ (d \text{ choose } k)\ \gamma(n+k)\ X^k.$$

This construction keeps only d+1 consecutive coefficients, yet it preserves enough structure to pose a sharp geometric question: are all d roots real? In the normalization used here the desired conclusion is stronger. After the final change of variables, the roots should be real, distinct and negative.

A polynomial with only real roots is called hyperbolic. Jensen and George Pólya showed that sequences of hyperbolic Jensen polynomials are closely related to entire functions in the Laguerre-Pólya class. This does not turn any one finite computation into a statement about all zeros of xi. It does explain why these polynomials are a natural finite probe of an infinite function.

What the theorem says. *There is an effective constant K such that the degree-d Jensen polynomial built from the nth xi coefficient has d distinct negative roots whenever n and d lie in a specified two-thirds wedge. In the headline asymptotic form, $n^2 \log(n+2)$ is at least Kd^3.*

What it does not say. *It does not prove that every xi Jensen polynomial is hyperbolic, and it does not prove that every nontrivial zero of zeta lies* *on the critical line. The effective constant is not optimized and the*

resulting cutoff is far beyond the range of ordinary numerical testing.

Fixed degree, growing degree

The modern study of xi's Jensen polynomials changed markedly with work by Michael Griffin, Ken Ono, Larry Rolen and Don Zagier. They proved that for each fixed degree d, the suitably shifted and rescaled Jensen polynomials approach the Hermite polynomial of that degree as n tends to infinity. Hermite polynomials have simple real roots, so the xi Jensen polynomials are eventually hyperbolic for every fixed d.

The phrase “for every fixed d” leaves open a uniform question. Suppose d is allowed to grow with n. How fast can it grow while one proof still controls all the polynomials in the region? A fixed-degree limit does not answer this: the error terms may deteriorate with d, and neighboring roots of a degree-d comparison polynomial become closer together as d increases.

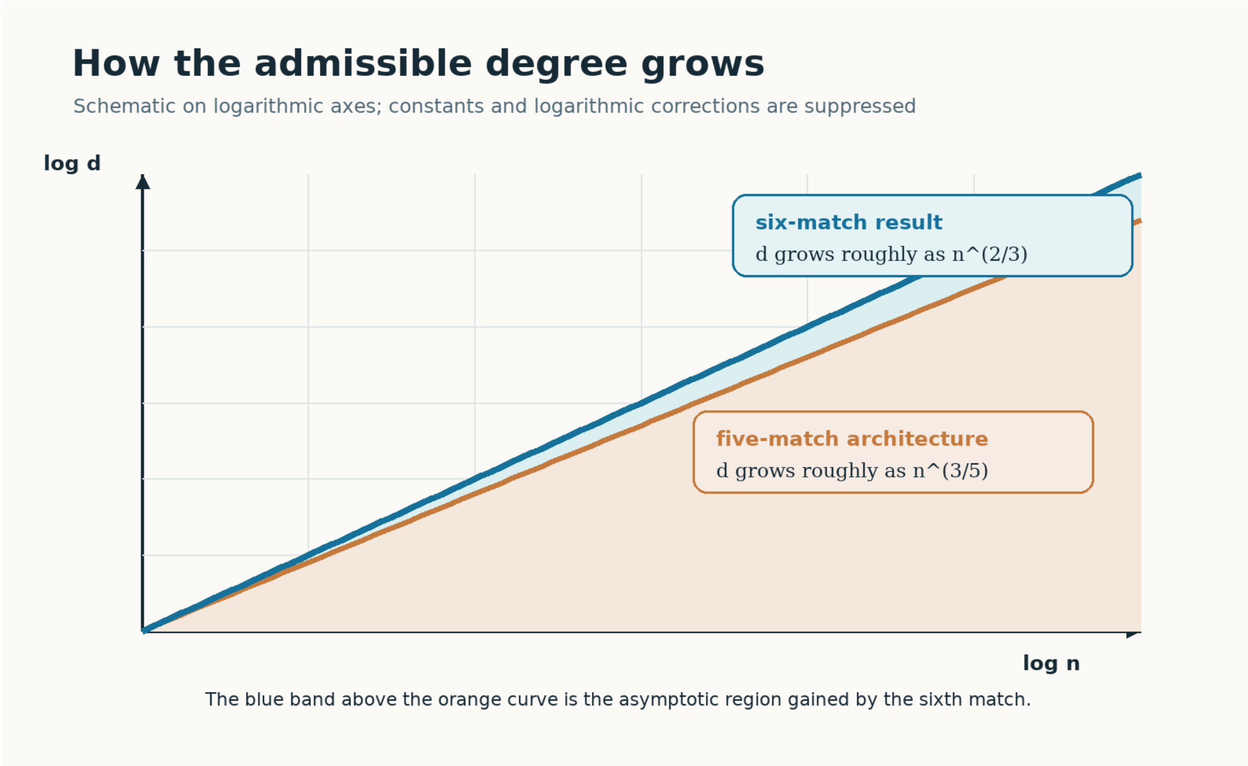
One way to picture a growing-degree theorem is to place log n on the horizontal axis and log d on the vertical axis. The allowed pairs occupy a wedge beneath a sloping boundary. An architecture developed by Jonathan Holland leads, after suppressing constants, to a condition of the form

$$n^3 \log^2(n+2) \geq Kd^5,$$

which permits d to grow roughly as $n^{3/5}$. The present candidate instead proves

$$n^2 \log(n+2) \geq Kd^3,$$

so d can grow roughly as $n^{2/3}$ times a mild logarithmic factor.



Schematic comparison of the three-fifths and two-thirds admissible wedges.

The diagram is deliberately schematic. It suppresses the different constants and logarithmic corrections and is not a plot of verified finite cases. Its point is the slope: two-thirds eventually dominates three-fifths. The extra region is not obtained by tightening a numerical constant. It requires one more order of cancellation in the comparison between xi and the model polynomial.

A model made from two Jacobi factors

The argument does not try to locate the roots of the xi Jensen polynomial directly. It constructs a comparison polynomial whose roots are easier to control and then proves that the xi polynomial is close enough to preserve their sign pattern.

The model is a terminating hypergeometric polynomial of the form

$$ {}_3F_2(-d, A, C; B, D; (D/AC)y). $$

It can be produced from two Jacobi-polynomial factors by a multiplicative finite-free convolution. Jacobi polynomials are classical orthogonal polynomials. In their natural parameter range their roots are real, simple and confined to a known interval. Finite-free convolution provides a way to combine those factors while retaining useful control over the product's root geometry.

Four positive parameters in the model are allowed to vary with n and d. Two normalizations already force the first two coefficient conditions; four equations then use the four parameters to enforce four more. In total, the normalized xi and model coefficients agree at six consecutive positions.

At the limiting point the parameter system has the exact positive solution

$$(\alpha, t, w, \delta) = (3, 2, 16/3, 1/3).$$

Its Jacobian determinant is exactly -1/144, and the infinity norm of the inverse Jacobian is 304/3. Those exact values matter because the proof needs more than a numerical solution at one point. A quantitative implicit-function argument must show that a unique positive solution persists throughout the prescribed parameter box for all sufficiently large n.

Why does the sixth match change the exponent? Write the normalized coefficient quotient as the exponential of a logarithmic error. If six node values agree, the lower-order portion of that error disappears. A Hermite-Genocchi or Newton remainder then expresses the first surviving term using a sixth derivative. Schematically, the comparison error changes from a fifth-order scale to a sixth-order scale. The root-separation cost grows with d, and balancing those two quantities produces the new cubic inequality $Kd^3 \leq n^2 \log n$.

This is the central structural claim of the work: the power two-thirds is the analytic and algebraic consequence of six matches, not an empirically fitted number.

The analytic engine inside γ(n)

Six coefficient matches are useful only if one has a uniform sixth-order description of the true xi coefficients. That is the more delicate analytic part of the argument.

Riemann's theta representation gives γ(n) as a Mellin-type integral of a rapidly decaying kernel. After the relevant substitution, the dominant exponential contains a term πe^u. For a large complex parameter N, its saddle L_N is determined by the equation

$$N = L\_N(\pi e^{L_N} + 3/4).$$

On the positive real axis this equation has a unique solution. The proof must do considerably more: it constructs a holomorphic saddle branch in a fixed complex sector, chooses the logarithm and square root consistently, and tracks quantitative bounds on the branch and its derivatives.

The contour of integration is then deformed through the saddle in the horizontal steepest-descent direction. Near L_N, the phase has a negative quadratic term, so the central contribution becomes Gaussian. Outside a small window, connector segments, remote tails and higher theta modes must all be shown uniformly smaller than the main contribution. The contour must stay inside the legal domain of the principal logarithm; otherwise a familiar formal Gaussian calculation would rest on an invalid deformation.

The paper organizes this analytic chain into five modules:

1. the exact Mellin normalization connecting the theta integral to γ(n);

2. the existence and control of the sectorial saddle branch;
3. the leading contour calculation, including its central Gaussian and tails;
4. the uniform suppression of every higher theta mode; and
5. the assembled sectorial coefficient theorem, on a domain slightly larger than the one ultimately used.

The larger domain in the fifth step permits Cauchy's integral formula to differentiate the asymptotic expansion without differentiating an informal big-O term. After the chain rule and the gamma normalization are handled consistently, the leading derivative constants from orders two through six are

> *2, -2, 4, -12, 48.*

At sixth order the exact rational function has denominator $(4+4r-3\sigma)^{12}$. Its reduced numerator has 82 terms and total degree 13. On the certified parameter box, a coefficientwise rational majorant is less than 10,000.

These fingerprints are useful because they can be independently recalculated. Recovering the same denominator power, term count, total degree, chain-rule factor 48 and exact majorant does not prove that the surrounding analysis is correct. It does, however, make several classes of algebraic and normalization error comparatively easy to detect.

Why six equal values control an interval

The sixth-order remainder is the bridge between local coefficient matching and uniform control.

Suppose a function F vanishes at six designated nodes. The divided-difference form of the Hermite-Genocchi formula represents F at another point as a sixfold product times an average of F's sixth derivative over a simplex. The simplex has mass $1/6! = 1/720$. This exact factorial is not cosmetic: losing it would destroy the numerical scale needed by the later comparison.

For the xi/model quotient, the six matches provide those six zeros. The analytic derivative theorem bounds the sixth derivative throughout a complex tube around the real interval. The result is a uniform sixth-order residual estimate, rather than six isolated equalities.

There are two ways this step can go wrong in a manuscript. One can quote a real-variable interpolation formula even though the nodes and intermediate points are complex; or one can state that six coefficient equalities are “equivalent” to the quotient-node conditions without displaying the normalizations. The supporting development now includes both adapters explicitly: a complex Hermite-Genocchi theorem and the algebraic identity that converts the four solved equations plus two normalizations into the six node values.

From a small multiplier to d real roots

Controlling coefficients is not yet controlling roots. A polynomial whose roots nearly collide can be sensitive to a small perturbation. The algebraic side of the proof therefore develops quantitative root separation for the model before transferring any conclusion to xi.

The Jacobi factors admit symmetric tridiagonal matrix representations. Matrix localization bounds place their roots in explicit intervals. A strict logarithmic-mesh theorem controls their multiplicative spacing, and a finite-free product-root inequality transports localization through the convolution. The hypergeometric differential equation supplies a recurrence for the model and its derivatives. A global maximum argument then rules out a first normalized derivative that exceeds the prescribed radius.

Now package the difference between the transformed xi polynomial and the model as a multiplier c_F. The six matches give $c_F = 1$ at the six nodes. The sixth-order residual theorem proves, uniformly on the interval containing the model's critical points,

$$|c_F - 1| < 1.$$

At successive critical points the model polynomial alternates sign. A multiplicative correction lying strictly inside the unit disk cannot erase those signs. The intermediate value theorem therefore supplies a root in each intervening interval. The intervals are disjoint, so the roots are distinct. Degree counting gives all d roots, and the final reciprocal and scale transformation places them on the negative real axis in the original xi Jensen coordinate.

The word “concrete” is important here. It is easy to formalize a general lemma saying that *if* some multiplier obeys the right node and interval conditions, roots are preserved. The latest stage goes further: it constructs the xi-specific multiplier, proves its six node values, derives its strict unit bound from the residual theorem, instantiates the interval certificate, and identifies the output with the actual transformed xi Jensen polynomial.

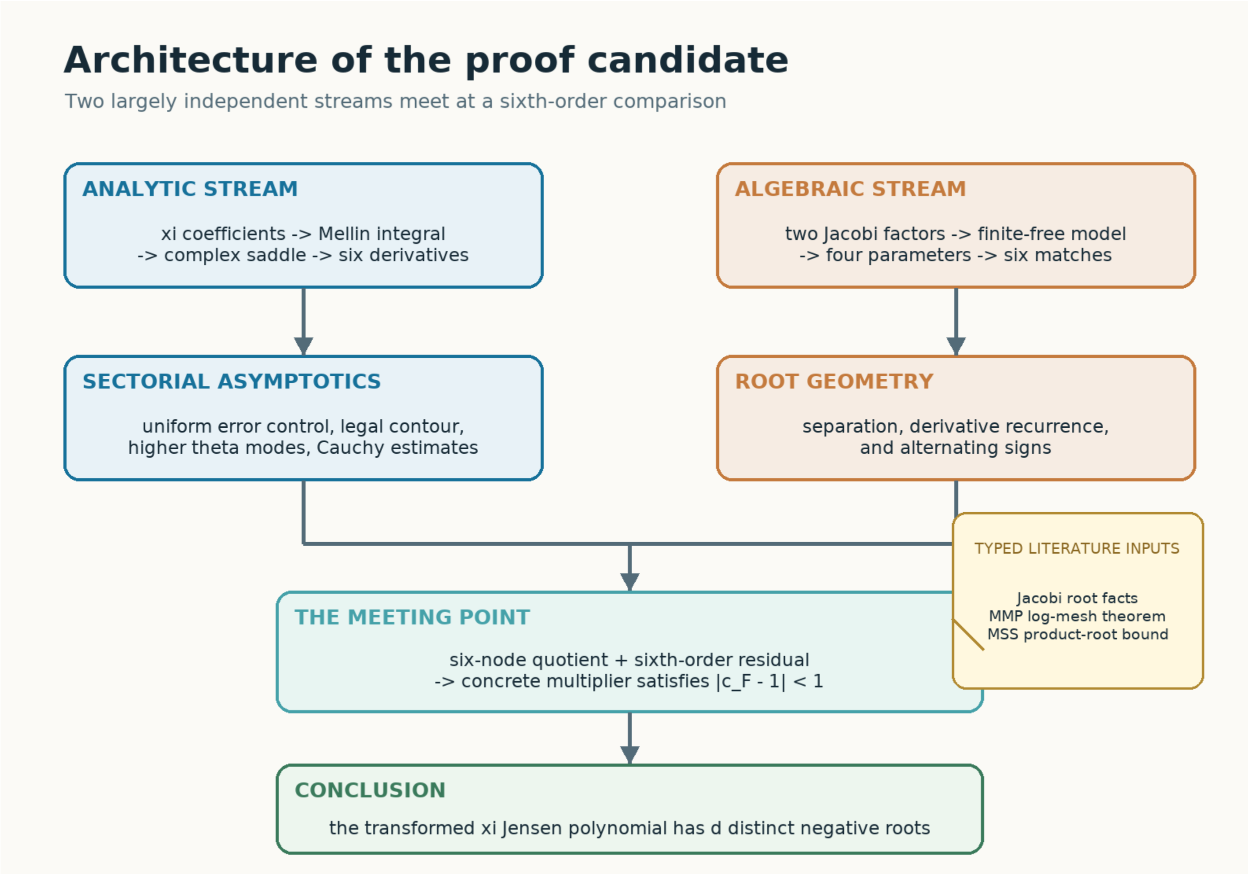


Diagram of the analytic and algebraic proof streams, their sixth-order meeting point, and the typed literature inputs.

What the latest Lean theorem adds

Lean is a language in which mathematical objects, hypotheses and conclusions are written precisely enough for a small kernel to verify every logical inference. A Lean proof is valuable only to the extent that its formal statement matches the intended mathematics and its assumptions are visible.

Earlier formal stages established much of the analytic and algebraic infrastructure but stopped before the final xi-specific sixth-multiplier certificate. That endpoint left a reasonable question: had the generic stability mechanism actually been connected to the polynomial named in the paper?

The newest development closes that seam. Its terminal declaration, `riemannXiJensen_twoThirds_global_headline_exactly`, builds the concrete multiplier certificate, performs the analytic-range versus finite-cutoff split, and concludes that the xi Jensen polynomial has exactly the expected number of distinct negative roots. Lean separately proves that the polynomial has degree exactly `d`, so “exactly” is not being inferred from informal degree counting outside the theorem.

The theorem does not import the entire mathematical literature into Lean. It takes one structured value containing three families of classical root facts:

- the required Jacobi root and matrix properties;
- the strict log-mesh preservation theorem associated with Martínez-Finkelshtein, Morales and Perales; and
- the finite-free product-root inequality associated with Marcus, Spielman and Srivastava.

Those are ordinary, explicitly typed hypotheses. They are not hidden project axioms. The terminal axiom audit reports only Lean's standard logical principles, and automated scans reject `sorry`, `admit`, custom axioms, unsafe escape hatches and equivalent shortcuts.

> **The formal boundary.** *Lean checks the new xi-specific analytic, sixth-order, multiplier and sign-transfer assembly. The three cited literature inputs above remain literature inputs; their original proofs have not been reimplemented from first principles. The typed MSS interface requires positive-root and degree certificates for both factors and strict positivity of both interval lower endpoints; Lean proves the concrete lower-endpoint inequalities before invoking that external result.*

This boundary is a feature, not a disclaimer to be minimized. A reader can see exactly which statements are checked by the kernel and exactly which are being accepted from cited work. That is more informative than the undivided label “formally verified.”

Independent algebra and interval tripwires

Formal proof and computer algebra address different risks. Lean is well suited to logical composition and explicit dependencies, but it is not the best environment in which to discover or independently reconstruct an 82-term rational function. Computer-algebra systems can do that efficiently, but a successful symbolic simplification does not certify the analytic domain on which an identity is used.

The project therefore uses a layered verification record.

SymPy scripts build the parameter equations, derivative tower, hypergeometric recurrence and interval polynomials from their definitions. A separate clean-room Mathematica notebook was executed without importing the repository's frozen expressions, JSON records, Lean terms or SymPy outputs. It recovered the positive parameter solution, determinant, inverse norm, shifted recurrence, sixth-derivative denominator, 82-term numerator and exact majorant. This is a second-computer-algebra-system recalculation, not an independent human derivation: the notebook cells themselves were prepared with AI assistance and executed by the author.

Directed Arb/ACB computations add rigorous enclosures for selected complex boxes, contour quantities and polynomial roots. Exact rational interval checks verify the margins in the branch boxes and parameter inequalities. Numerical evaluations of the Mellin representation and its saddle approximation test the normalization across widely separated scales.

Finally, semantic mutation tests deliberately alter load-bearing statements: a factor of eight, a strict inequality, the saddle phase, a recurrence coefficient, the multiplier alias or its connection to xi. The verification gate must fail on each mutation. Mutation testing is not a proof of the theorem. It is evidence that the tests are attached to the claimed seam rather than merely recognizing filenames or reassuring phrases.

What automated scrutiny has already changed

The review archive is intentionally not a sequence of unbroken passes. AI-only adversarial reviews found substantive defects in earlier drafts. One version printed an incorrect coefficient integral. Another misstated the critical-radius hypothesis used in sign transfer. A later review noticed that a contour described as horizontal had been parameterized in the imaginary direction, changing Gaussian descent into ascent. Another caught a mismatch between the logarithm of the Mellin moment and the logarithm of the normalized coefficient.

Those were not stylistic objections. Each could have invalidated the printed argument at that point. The formulas were repaired, the affected reasoning was rechecked, and targeted regressions were added. Subsequent review also forced the contour's logarithmic branch cut, the definition of the phase and amplitude, and the identity of the multiplier polynomial to be stated rather than inferred.

A fresh clean-packet AI review then found a formal defect of a different kind. The MSS input record quantified over arbitrary lower endpoints. At positive degree, choosing negative symmetric endpoints made that record inconsistent, so the headline implication could be satisfied only vacuously even though the paper's intended application used positive intervals. The interface now carries the missing positive-root, degree and strict-endpoint conditions, and Lean derives the two endpoint inequalities from `B,D ≥ 256d`. A regression test preserves the old call and requires Lean to reject it at the missing positivity argument.

This history does not make the present candidate infallible. AI reviewers can share blind spots, especially when trained on similar mathematical language or supplied with the same evidence. Its value is narrower: it records real attempts to falsify the argument and shows how the proof surface changed in response. The repaired candidate has passed the project's Lean, source, mutation and reproducibility gates; a fresh independent AI re-review of that candidate remains pending.

What a wider wedge would mean

If the proof survives expert scrutiny, it establishes a larger uniform real-rootedness region for the xi Jensen polynomials. It also demonstrates that the Jacobi finite-free comparison architecture can be pushed one order beyond its five-match form. The analytic work shows that the required sixth derivative remains uniformly controlled in a complex sector; the algebraic work shows how that derivative control reaches actual roots.

The exponent should not be interpreted as a universal barrier. It describes what six matches buy within this architecture. A seventh match would require additional parameters or a new relation among the existing ones. Optimizing the effective constant would be a different project: many present estimates were chosen for transparency and uniformity rather than numerical sharpness.

Nor is the theorem a practical algorithm for checking small Jensen polynomials. Direct numerical root finding works far below the asymptotic cutoff. The theorem's purpose is to explain a uniform regime as n and d grow together.

The broader methodological point may be equally interesting. Formal proof is often most persuasive not when it presents a monolithic seal of correctness, but when it makes the architecture inspectable. Here one can move from the coefficient integral to the saddle theorem, from the saddle theorem to six derivatives, from six matches to a residual, and from the residual to the actual xi polynomial. One can also see where the chain leaves the formal development and invokes established literature.

That kind of transparency is especially important for research conducted outside an academic institution and developed with substantial AI assistance. Software can make a long argument easier to explore, formalize and test. It can also make an incorrect argument look finished. The proper response is not to treat computation as a substitute for mathematical judgment, but to expose enough structure that knowledgeable readers can test the claims at their most vulnerable seams.

Questions for a critical reader

The most useful response to this work is not a general vote of confidence. It is a precise challenge. Among the questions the release is designed to make answerable are:

- Does the Mellin normalization reproduce the stated xi coefficients?
- Is the complex saddle branch controlled on the full sector used by Cauchy's formula?
- Does the contour deformation remain clear of the principal-logarithm cut?
- Are all higher theta modes bounded uniformly rather than sampled?
- Do the four parameter equations plus two normalizations give exactly the six quotient-node values?
- Does the complex sixth-order remainder carry the correct 1/720 mass?

- Are the hypotheses of the cited Jacobi, log-mesh and product-root theorems satisfied in the chosen parameter range?
- Is the strict multiplier bound connected definitionally to the actual transformed xi Jensen polynomial?
- Does the prose theorem have the same quantifiers and cutoff as the formal headline theorem?

The paper, supplement, formal source, clean-room notebook and mutation record are supplied so that these questions can be investigated at the level of definitions rather than slogans.

Read and inspect the work

- **Full paper:** included with this article
- **Permanent Version 1.0 record and DOI:** supplied with this article
- **Public source and immutable release:** supplied with this article
- **Technical supplement and complete reviewer package:** included with the permanent record

Disclosure: AI systems assisted substantially with the research, formalization, software, writing, and AI-only adversarial review. No human mathematical review or peer review is claimed. The work does not prove the Riemann hypothesis.